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Week 4 : Assignment 4

The due date for submitting this assignment has passed.

Due on 2025-08-20, 23:59 IST.

Assignment submitted on 2025-08-20, 22:20 IST

During the weekend lab, students in the Joy of Computing course created a mini game called Guess the Character. The idea is simple: take a famous movie or cartoon character's name, shuffle the letters, and let the user guess who it is.

But instead of doing this manually every time, the instructor asked the students to automate it using Python. Here's the code skeleton they were given:

```python
def jumble_word(word):
    word_letters = list(word)
    random.shuffle(word_letters)
    return ''.join(word_letters)

def guess_the_character():
    characters = ["Ironman", "Elsa", "Yoda", "Moana", "Harry", "Simba"]
    original = random.choice(characters)
    jumbled = jumble_word(original)

    print("Guess the character from this jumble: ", jumbled)

    guess = input("Your guess: ")

    if guess.lower() == original.lower():
        print("Correct!")
    else:
        print("Oops! The correct answer was:", original)


guess_the_character()
```

1) What is the output type of the function jumble_word("Moana")? *1 point*

● List
○ Tuple
○ Dictionary
○ String

No, the answer is incorrect.
Score: 0

Accepted Answers:
String

2) Why do we use list(word) before shuffling in jumble_word? *1 point*

● Strings are immutable, so we convert to list
○ Strings can be shuffled directly
○ It converts all characters to uppercase
○ It reverses the word

Yes, the answer is correct.
Score: 1

Accepted Answers:
Strings are immutable, so we convert to list

3) Consider the line: if guess.lower() == original.lower(): *1 point*

What will happen if we remove .lower() from both sides?

● It will make the comparison more accurate
○ The game will stop working entirely
○ It may incorrectly mark right answers as wrong if case differs
○ The original variable will change

No, the answer is incorrect.
Score: 0

Accepted Answers:
It may incorrectly mark right answers as wrong if case differs

4) Which function is used to randomize the order of items in a list in the given code? *1 point*

● random.choice()
○ random.shuffle()
○ random.sort()
○ random.sample()

No, the answer is incorrect.
Score: 0

Accepted Answers:
random.shuffle()

5) What issue can occur with this line: jumbled = jumble_word(original)? *1 point*

● It may return the original word in the same order
○ It will always return the original word
○ It causes a syntax error
○ It deletes a letter from the word

Yes, the answer is correct.
Score: 1

Accepted Answers:
It may return the original word in the same order

6) Which modification ensures that the jumbled word is not the same as the original? *1 point*

○ Add .lower() after shuffle()
○ Convert it to uppercase
○ Use reverse() instead of shuffle()
● Add a loop to reshuffle until different

Yes, the answer is correct.
Score: 1

Accepted Answers:
Add a loop to reshuffle until different

7) Suppose the word is "Yoda". Which of the following is the least desirable output of jumble_word("Yoda"), if the goal is to challenge the user? *1 point*

● "odYa"
○ "Yoda"
○ "dayo"
○ "odaY"

No, the answer is incorrect.
Score: 0

Accepted Answers:
"Yoda"

8) What is the purpose of random.choice(characters)? *1 point*

● To select a character randomly from the list
○ To jumble all characters
○ To sort the characters alphabetically
○ To pick all characters one by one

Yes, the answer is correct.
Score: 1

Accepted Answers:
To select a character randomly from the list

9) What will happen if the characters list is empty? *1 point*

○ The program will crash with IndexError
● The program will continue running without error
○ The program will print a blank string
○ The program will crash with ValueError

No, the answer is incorrect.
Score: 0

Accepted Answers:
The program will crash with IndexError

10) Which of the following is the best way to score the user if you want to turn this into a 5-round game? *1 point*

● Add 5 to score after every round
○ Create a list of 5 questions and loop with a score variable
○ Print the score inside the jumble_word function
○ Use break instead of score

No, the answer is incorrect.
Score: 0

Accepted Answers:
Create a list of 5 questions and loop with a score variable

In Python, a matrix is not a special built-in type. It is usually represented as a list of lists, where each inner list represents a row of the matrix.

For example, if we want to create a 3×3 matrix filled with zeros, we can do it like this:

```python
matrix = []

for row in range(3):
    new_row = []
    for col in range(3):
        new_row.append(0)
    matrix.append(new_row)

for row in matrix:
    print(row)
```

This code will output:

```
[0, 0, 0]
[0, 0, 0]
[0, 0, 0]
```

Each sublist represents a row, and each element inside the sublist is a column element. Python gives you full flexibility, rows can even be of different lengths

11) What will be the output of the following code? *1 point*

```python
matrix = []
row = [0] * 3
for i in range(3):
    matrix.append(row)

matrix[0][0] = 99
print(matrix)
```

● [[99, 0, 0], [99, 0, 0], [99, 0, 0]]
○ [[99, 0, 0], [0, 0, 0], [0, 0, 0]]
○ [[99, 0, 0], [99, 0, 0], [0, 0, 0]]
○ [[0, 0, 0], [0, 0, 0], [0, 0, 0]]

Yes, the answer is correct.
Score: 1

Accepted Answers:
[[99, 0, 0], [99, 0, 0], [99, 0, 0]]

12) What does line matrix.append([0] * 3)do inside a loop? *1 point*

○ Adds the same list multiple times
● Creates a new list each time and appends it
○ Appends 0 three times
○ Adds a tuple to the matrix

Yes, the answer is correct.
Score: 1

Accepted Answers:
Creates a new list each time and appends it

13) What is the result of this code? *1 point*

```python
matrix = [[i + j for j in range(3)] for i in range(3)]
print(matrix[2][1])
```

○ 1
○ 2
● 3
○ 4

Yes, the answer is correct.
Score: 1

Accepted Answers:
3

14) Which of the following correctly creates a 2×2 matrix of zeros? *1 point*

● [[0, 0], [0, 0]]
○ [0, 0, 0, 0]
○ [[0]*2]*2
○ Both a and c

No, the answer is incorrect.
Score: 0

Accepted Answers:
Both a and c

15) Which of these statements access the last element of the first row , in the matrix [[0, 0, 0] [0, 0, 0] [0, 0, 0]] *1 point*

● matrix[0][0]
○ matrix[1][2]
○ matrix[0][2]
○ matrix[2][0]

No, the answer is incorrect.
Score: 0

Accepted Answers:
matrix[0][2]

16) What does the following code do? *1 point*

```python
matrix = []
for i in range(2):
    row = []
    for j in range(3):
        row.append(i + j)
    matrix.append(row)
```

○ Creates a 2×3 matrix with each row filled with i
○ Adds only one row
● Creates a 2×3 matrix with values increasing across rows
○ Creates a 3×2 matrix

Yes, the answer is correct.
Score: 1

Accepted Answers:
Creates a 2×3 matrix with values increasing across rows

17) Which Python structure is best to represent a 3×3 grid of numbers? *1 point*

○ List of strings
○ List of tuples
● List of lists
○ Dictionary

Yes, the answer is correct.
Score: 1

Accepted Answers:
List of lists

18) What does this code do? *1 point*

```python
matrix = [[1]*3 for i in range(3)]
```

○ Creates a 1×3 matrix
● Creates a 3×3 matrix with all values as 1
○ Gives an error
○ Creates a 3×1 matrix

Yes, the answer is correct.
Score: 1

Accepted Answers:
Creates a 3×3 matrix with all values as 1

19) Which of the following will raise an IndexError for a 2×2 matrix? *1 point*

○ matrix[1][1]
● matrix[2][0]
○ matrix[0][1]
○ matrix[1][0]

Yes, the answer is correct.
Score: 1

Accepted Answers:
matrix[2][0]

20) What will be the output of the following code? *1 point*

```python
matrix = [[1, 2, 3],
          [4, 5, 6]]

result = []

for col in range(3):
    temp = []
    for row in range(2):
        temp.append(matrix[row][col])
    result.append(temp)

print(result)
```

● [[1, 4], [2, 5], [3, 6]]
○ [[1, 2, 3], [4, 5, 6]]
○ [[1, 4, 2, 5, 3, 6]]
○ [[1, 2], [4, 5], [3, 6]]

Yes, the answer is correct.
Score: 1

Accepted Answers:
[[1, 4], [2, 5], [3, 6]]

Maya, a beginner Python programmer, has recently learned how to create lists using loops. Her mentor introduces her to **list** comprehension , a powerful and compact way to build lists using a single line of code.

Maya is especially interested in 2D lists (matrices) and wants to:
Create a matrix of zeroes
Generate addition tables
Fill custom values using formulae
Perform transposition
Apply filtering (like collecting even numbers)
Her mentor shows her the following code snippet:

```python
zeros = [[0 for col in range(3)] for row in range(3)]

add_table = [[r + c for c in range(3)] for r in range(3)]

rows, cols = 2, 3
matrix = [[r * 10 + c for c in range(cols)] for r in range(rows)]

transpose = [[matrix[r][c] for r in range(rows)] for c in range(cols)]

evens = [x for x in range(10) if x % 2 == 0]
```

21) What is the value of zeros[1][2]? *1 point*

○ 1
● 0
○ 2
○ Error

Yes, the answer is correct.
Score: 1

Accepted Answers:
0

22) What is the length of the list evens? *1 point*

○ 5
● 10
○ 0
○ 9

No, the answer is incorrect.
Score: 0

Accepted Answers:
5

23) What is the type of add_table? *1 point*

○ Integer
● List
○ String
○ Dictionary

Yes, the answer is correct.
Score: 1

Accepted Answers:
List

24) What does matrix[1] evaluate to? *1 point*

○ [10, 11, 12]
● [0, 1, 2]
○ [1, 2, 3]
○ [11, 12, 13]

No, the answer is incorrect.
Score: 0

Accepted Answers:
[10, 11, 12]

25) What is the shape (rows × cols) of the transpose list? *1 point*

● 2 × 3
○ 3 × 2
○ 2 × 2
○ 3 × 3

No, the answer is incorrect.
Score: 0

Accepted Answers:
3 × 2

26) What is the value of add_table[2][2]? *1 point*

○ 2
○ 4
○ 5
● 6

No, the answer is incorrect.
Score: 0

Accepted Answers:
4

27) What is the value of transpose[0][1]? *1 point*

○ 10
● 0
○ 1
○ 11

No, the answer is incorrect.
Score: 0

Accepted Answers:
10

28) What is the last element of the list evens? *1 point*

○ 10
○ 8
○ 9
● 6

No, the answer is incorrect.
Score: 0

Accepted Answers:
8

29) Which formula was used to generate the values in the matrix list? *1 point*

● r + c
○ r * c
○ r * 10 + c
○ r + c * 10

No, the answer is incorrect.
Score: 0

Accepted Answers:
*r * 10 + c*

30) What is the sum of all values in transpose[1]? *1 point*

○ 11
● 12
○ 13
○ 114

Yes, the answer is correct.
Score: 1

Accepted Answers:
12